Isometry Pursuit: Convex Recovery of Orthonormal Column Submatrices

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Paper under double-blind review

Abstract

Isometry pursuit is a convex algorithm for identifying orthonormal column-submatrices of wide matrices. It consists of a vector normalization followed by multitask basis pursuit. Applied to Jacobians of putative coordinate functions, it helps identity locally isometric embeddings from within interpretable dictionaries. We provide theoretical and experimental results justifying this method, including a proof with realistic assumptions that such isometric submatrices, should they exist, are contained within the obtained support. For problems involving coordinate selection and diversification, it offers a synergistic alternative to greedy and brute force search.

1 Introduction

Many real-world problems may be abstracted as selecting a subset of the columns of a matrix representing stochastic observations or analytically exact data. This paper focuses on a simple such problem. Given a rank D matrix $X \in \mathbb{R}^{D\times P}$ with $P > D$, select a square submatrix $X_{\cdot S}$ where subset $S \subset [P]$ satisfies $|S| = D$ that is as isometric as possible.

This problem arises in interpretable learning specifically because while the coordinate functions of a given feature space may have no intrinsic meaning, it is sometimes possible to generate a dictionary of interpretable features for consideration as potential parametrizing coordinates. When this is the case, selection of candidate interpretable features as coordinates can take the above form. While implementations vary across data and algorithmic domains, identification of such coordinates generally aids mechanistic understanding, generative control, and statistical efficiency.

This paper shows that an adapted version of the algorithm in **?** leads to a convex alternative to the greedy approaches such as those in **????** for finding isometries. This algorithm - IsometryPursuit - is multitask basis pursuit applied to an appropriately normalized X selects orthonormal submatrices. Given vectors in $\mathbb{R}^D$, the normalization log-symmetrizes length and favors those closer to unit length, while basis pursuit favors those which are orthogonal. Our theoretical results formalize this intuition, and show the usefulness of IsometryPursuit as a trimming procedure prior to brute force search for diversification and interpretable coordinate selection.

2 Background

Our algorithm is motivated by spectral and convex analysis.

2.1 Problem

Our goal is, given a matrix $X \in \mathbb{R}^{D\times P}$, to select a subset $S \subset [P]$ with $|S| = D$ such that $X_{\cdot S}$ is as orthonormal as possible in a computationally efficient way. To this end, we define a ground truth loss function that measures orthonormalness, and then introduce a surrogate loss function that convexifies the problem so that it may be efficiently solved. Figure 1a and 1b provide a visual depiction of this problem when $D = 2$ and $P = 30$.

2.2 Interpretability and isometry

Our motivating example is the selection of data representations from within sets of putative coordinates: the columns of a provided wide matrix. Compared with Sparse PCA (**????**), we seek a low-dimensional representation from the set of these column vectors rather than their span.

This method applies to interpretability, for which parsimony is at a premium. Interpretability arises through comparison of data with what is known to be important in the domain of the problem. This knowledge often takes the form of a functional dictionary. Evaluation of independence of dictionary features arises in numerous scenarios (**???**). The requirement that dictionary features be full rank has been called functional independence (**?**) or feature decomposability (**?**), with connection between dictionary rank and independence via the implicit function theorem. Besides independence, the metric properties of such dictionary elements are of natural interest. This is formalized through the notion of differential.

Definition 1 *The* ***differential*** *of a smooth map $\phi : \mathcal{M} \to \mathcal{N}$ between D dimensional manifolds $\mathcal{M} \subseteq \mathbb{R}^B$ and $\mathcal{N} \subseteq \mathbb{R}^P$ is a map in tangent bases $x_1 \dots x_D$ of $T_\xi \mathcal{M}$ and $y_1 \dots y_D$ of $T_{\phi(\xi)}\mathcal{N}$ consisting of entries*

$$D\phi(\xi) = \begin{bmatrix} \frac{\partial \phi_1}{\partial x_1}(\xi) & \dots & \frac{\partial \phi_1}{\partial x_D}(\xi) \\ \vdots & & \vdots \\ \frac{\partial \phi_D}{\partial x_1}(\xi) & \dots & \frac{\partial \phi_D}{\partial x_D}(\xi) \end{bmatrix}. \tag{1}$$

It is not always necessary to explicitly estimate tangent spaces when applying this definition. The most commonly encountered manifolds are vector spaces for which the tangent spaces are trivial. This is the case for full-rank tabular data, for which isometry has a natural interpretation as a type of diversification, and often for the latent spaces of deep learning models. In this case, $B = D$.

Definition 2 *A map ϕ between D dimensional submanifolds with inherited Euclidean metric $\mathcal{M} \subseteq R^B$ and $\mathcal{N} \subseteq R^P$ ϕ is an* ***isometry at a point*** *$\xi \in \mathcal{M}$ if*

$$D\phi(\xi)^T D\phi(\xi) = I_D. \tag{2}$$

That is, ϕ is an isometry at ξ if $D\phi(\xi)$ is orthonormal.

The applications of pointwise isometry are themselves manifold. Pointwise isometric embeddings faithfully preserve high-dimensional geometry. For example, Local Tangent Space Alignment (**?**), Multidimensional Scaling (**?**) and Isomap (**?**) non-parametrically estimate embeddings that are as isometric as possible. Another approach stitches together pointwise isometries selected from a dictionary to form global embeddings (**?**). The method is particularly relevant since it constructs such isometries through greedy search, with putative dictionary features added one at a time.

That $D\phi$ is orthonormal has several equivalent formulations. The one motivating our ground truth loss function comes from spectral analysis.

Proposition 1 *The singular values $\sigma_1 \dots \sigma_D$ are equal to 1 if and only if $U \in \mathbb{R}^{D \times D}$ is orthonormal.*

On the other hand, the formulation that motivates our convex approach is that orthonormal matrices consist of D coordinate features whose gradients are orthogonal and of unit length.

Proposition 2 *The component vectors $u_1 \dots u_D \in \mathbb{R}^B$ form a orthonormal matrix if and only if, for all $d_1, d_2 \in [D]$, $\langle u_{d_1}, u_{d_2} \rangle = \begin{cases} 1 & d_1 = d_2 \\ 0 & d_1 \neq d_2 \end{cases}$.*

2.3 Best subset selection

Given a matrix $X \in \mathbb{R}^{D\times P}$, we compare algorithmic paradigms for solving problems of the form

$$\arg\min_{S\in\binom{[P]}{D}} \ell(X_{\cdot S}) \tag{3}$$

where $\binom{[P]}{D} = \{A \subseteq [P] : |A| = D\}$. Brute force algorithms consider all possible solutions. These algorithms are conceptually simple, but have the often prohibitive time complexity $O(C_\ell P^D)$ where C_ℓ is the cost of evaluating ℓ. Greedy algorithms consist of iteratively adding one element at a time to S. This algorithms have time complexity $O(C_\ell PD)$ and so are computationally more efficient than brute force algorithms, but can get stuck in local minima. Formal definitions are given in Section 4.1.

Sometimes, it is possible to introduce an objective which convexifies problems of the above form (**???**). Solutions

$$\arg\min f(\beta) : Y = X\beta \tag{4}$$

to the overcomplete regression problem $Y = X\beta$ are a classic example (**?**). When $f(\beta) = \|\beta\|_0$, this problem is non-convex, and is thus suitable for greedy or brute algorithms, but when $f(\beta) = \|\beta\|_1$, the problem is convex, and may be solved efficiently via interior-point methods. When the equality constraint is relaxed, Lagrangian duality may be used to reformulate as a so-called Lasso problem, which leads to an even richer set of optimization algorithms (**?**).

The form of basis pursuit that we apply is inspired by the group basis pursuit (**??**). Multitask basis pursuit is a special form of group basis pursuit which we call differently since grouping is dependent only on the structure of matrix-valued response variable y. The objective $f(\beta) = \|\beta\|_{1,2} := \sum_{p=1}^{P} \|\beta_{p\cdot}\|_2$ creates joint sparsity across entire rows $\beta_{p\cdot}$ (**???**). It was used in **?** to select independent subsets among interpretable features.

3 Method

We apply the multitask basis pursuit paradigm used to select independent dictionary elements in **??** to the more specific problem of selecting pointwise isometries from a dictionary. We first define a ground truth objective computable via brute and greedy algorithms that is uniquely minimized by orthonormal matrices. We then define the combination of normalization and multitask basis pursuit that approximates this ground truth loss function. We finally give a brute post-processing method for ensuring that the solution is D sparse, and provide a theoretical result that the two stage approach will always result in recovery of an isometric solution from the dictionary, should one exist.

3.1 Ground truth

We'd like a ground truth objective to be minimized uniquely by orthonormal matrices, invariant under rotation, and depend on all changes in the matrix. Deformation (**?**) and nuclear norm (**?**) use only a subset of the differential's information and are not uniquely minimized at unitarity, respectively. We therefore introduce an alternative ground truth objective that satisfies the above desiderata and has convenient connections to isometry pursuit.

This ground truth objective is

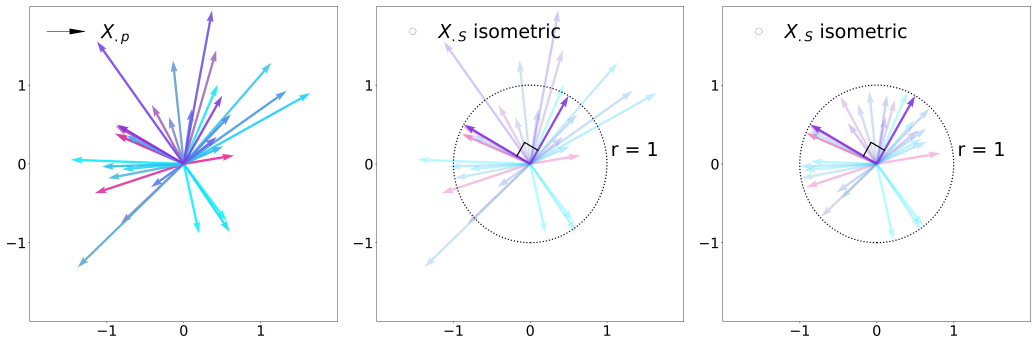
$$\ell : \mathbb{R}^{D\times P} \to \mathbb{R}^+ \tag{5}$$

$$X \mapsto \sum_{d=1}^{D} g(\sigma_d(X)) \tag{6}$$

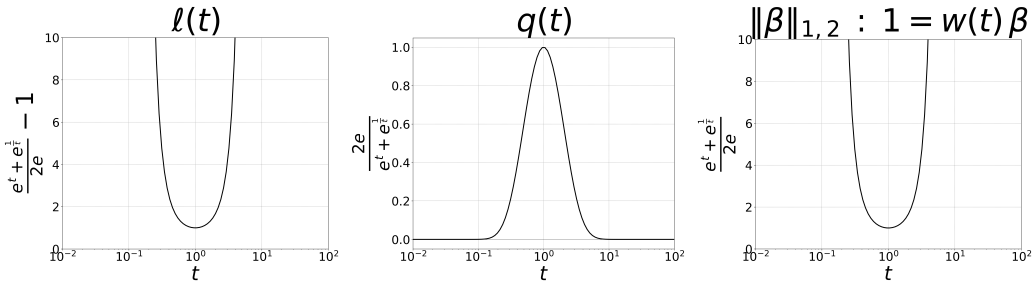
where $\sigma_d(X)$ is the d-th singular value of X and

$$g : \mathbb{R}^+ \to \mathbb{R}^+ \tag{7}$$

$$t \mapsto \frac{e^t + e^{t^{-1}}}{2e}. \tag{8}$$



(a) Raw data (b) Isometric subset (c) Normalization step



(d) Ground truth loss (e) Normalized length vs. unnormalized length (f) Basis pursuit loss

Figure 1: 1a Raw data X represented as column-vectors. 1b An isometric subset - the identification of which is our objective. 1c The vectors after normalization so that vectors of length 1 maintain longest length. 1d Our ground truth loss function in the one-dimensional case $D = 1$ 1e Normalized length used to rescale Figure 1a to get vectors in Figure 1c. 1f Basis pursuit loss. This loss is equivalent to that shown in Figure 1d in the one-dimensional case.

By Proposition 1, we can see that ℓ is uniquely maximized by orthonormal matrices. Moreover, g is convex, and $\ell(X^{-1}) = \ell(X)$ when X is invertible. Figure 1d gives a graph of ℓ when $D = 1$.

Our ground truth objective is therefore the best subset selection problem

$$\arg \min_{S \in \binom{[P]}{d}} \ell(X_{\cdot S}). \tag{9}$$

Regardless of the convexity of ℓ, brute combinatorial search over $[P]$ is inherently non-convex. This motivates our alternative formulation.

3.2 Normalization

Since basis pursuit methods tend to select longer vectors, selection of orthonormal submatrices requires normalization such that both long and short candidate basis vectors are penalized in the subsequent regression. We introduce the following definition.

Definition 3 (Symmetric normalization) *A function $q : \mathbb{R}^D \to \mathbb{R}^+$ is a symmetric normalization if*

$$\arg \max_{v \in \mathbb{R}^D} \; q(v) = \{v : \|v\|_2 = 1\} \tag{10}$$

$$q(v) = q(\frac{v}{\|v\|_2^2}) \tag{11}$$

$$q(v_1) = q(v_2) \; \forall \; v_1, v_2 \in \mathbb{R}^D : \|v_1\|_2 = \|v_2\|_2. \tag{12}$$

We use such functions to normalize column-vector length in such a way that vectors of length 1 prior to normalization have longest length after normalization and vectors are shrunk proportionately to their deviation from 1. That is, we normalize vectors by

$$n : \mathbb{R}^D \to \mathbb{R}^D \tag{13}$$

$$v \mapsto q(v)v \tag{14}$$

and matrices by

$$w : \mathbb{R}^{D\times P} \to \mathbb{R}^D \tag{15}$$

$$X_{.p} \mapsto n(X_{.p}) \; \forall \; p \in [P]. \tag{16}$$

We choose q as follows.

$$q : \mathbb{R}^D \to \mathbb{R}^+ \tag{17}$$

$$v \mapsto \frac{e^{\|v\|_2} + e^{\|v\|_2^{-1}}}{2e}. \tag{18}$$

Besides satisfying the conditions in Definition 3, this normalization has some additional nice properties. First, q is convex. Second, it grows asymptotically log-linearly. Third, while $\exp(-|\log t|) = \exp(-\max(t, 1/t))$ is a seemingly natural choice for normalization, it is non smooth, and the LogSumExp (**?**) replacement of $\max(t, 1/t)$ with $\log(\exp(t) + \exp(1/t))$ simplifies to 18 upon exponentiation. This normalization provides good numerical properties for avoiding issues arising close to 0 and ∞. We plot this normalization function in Figure 1e and its effect on our example data in Figure 1c.

3.3 Isometry pursuit

Isometry pursuit is the application of multitask basis pursuit to the normalized design matrix $w(X)$ to identify submatrices of X that are as orthonormal as possible. Define the multitask basis pursuit penalty

$$\|\cdot\|_{1,2} : \mathbb{R}^{P\times D} \to \mathbb{R}^+ \tag{19}$$

$$\beta \mapsto \sum_{p=1}^{P} \|\beta_{p.}\|_2. \tag{20}$$

Given a matrix $Y \in \mathbb{R}^{D\times D}$, the multitask basis pursuit solutions are

$$\widehat{\beta}_{MBP}(X, Y) := \arg \min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \; : \; Y = X\beta. \tag{21}$$

The multitask basis pursuit solution is often not unique. See **?** for more details on uniqueness of basis pursuit solutions, for which the uniqueness condition is less strict than in the multitask case. We therefore define

$$\widehat{\beta}^{\ell_2}_{MLP}(X, Y) := \arg\min \|\beta\|_F : \beta \in \beta_{MBP}(X, Y). \tag{22}$$

By the strong convexity of the ℓ_2 norm, $\widehat{\beta}^{\ell_2}_{MLP}(X, Y)$ is well defined.

This implicit regularization assumption is reasonable. For example, in **?** the implicit selection of the minimum ℓ_2 norm solution among the lasso solutions was proven for the Least Angle Regression Selection (LARS) algorithm for solving the lasso, while **?** assumes that the group lasso solution is in fact min-norm prior to subsequent theoretical analyses. Empirically, the Splitting Conic Solver type method (which is itself a variant of the Alternating Direction Method of Multilpliers) (**?**) stably estimates the min-norm solution when initialized at 0, but besides a brief exploration in Section **??** we leave theoretical and experimental proof of this feature of the feature our optimization approach for future work, and instead make the ability of our optimizer to obtain the min-norm solution an assumption of our eventual proposition.

Isometry pursuit is then given by

$$\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) \tag{23}$$

where I_D is the D dimensional identity matrix and recovered functions are the indices of the dictionary elements with non-zero coefficients. That is, they are given by $S(\beta)$ where

$$S : \mathbb{R}^{P\times D} \to \binom{[P]}{D} \tag{24}$$

$$\beta \mapsto \{p \in [P] : \|\beta_{p\cdot}\| > 0\}. \tag{25}$$

IsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: Normalize $X' = w(X)$
2: Optimize $\widehat{\beta} = \widehat{\beta}^{\ell_2}_{MBP}(X', I_D)$
3: **Output** $\widehat{S} = S(\widehat{\beta})$

3.4 Theory

The intuition behind our application of multitask basis pursuit is that submatrices consisting of vectors which are closer to 1 in length and more orthogonal will have smaller loss. A key theoretical assertion is that IsometryPursuit is invariant to choice of basis for X.

Proposition 3 *Let $U \in \mathbb{R}^{D\times D}$ be orthonormal. Then $S(\widehat{\beta}(UX)) = S(\widehat{\beta}(X))$.*

This proposition establishes the basis-independence of Isometry Pursuit and that we may replace I_D in the constraint by any orthonormal $D \times D$ matrix. It is also essential for showing the following main theoretical result.

When a rank D orthonormal column-submatrix $X_{\cdot S}$ exists, the output of Program 23 will contain S.

Proposition 4 *Let $X \in \mathbb{R}^{D\times P}$ have a rank D orthonormal column submatrix $X_{\cdot S}$. Then $S \subseteq \widehat{\beta}(X)$.*

Proofs of these propositions are given in Section 4.2, with corresponding experimental results in Section **??**. The proof mostly relies on the KKT conditions (**?**) being used to verify that an isometric subset yields a corresponding basis pursuit solution for an appropriate normalized matrix. This is a formalization of our intuition that longer and more orthogonal vectors will minimize loss. We then apply a simple argument to show that the ℓ_2 minimizing solution is the least sparse solution, which means that it contains the isometry solution, should it exist.

3.5 Two-stage isometry pursuit

Proposition 4 suggests the following algorithm, which first uses the convex IsometryPursuit algorithm to prune the P candidate features and then applies brute search upon the reduced feature set. The proposition says that this approach is guaranteed to select an isometric submatrix should one exist. The two-stage approach is standard in the Lasso literature (**???**), and Proposition 4 justifies it in our application. This method forms our practical isometry estimator, the core advantage of which over purely brute search is a substantially improved computational complexity.

TwoStageIsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: $\widehat{S}_{IP} =$ IsometryPursuit(X)
2: $\widehat{S} =$ BruteSearch$(X_{\cdot \widehat{S}_{IP}}, \ell)$
3: **Output** $\widehat{S}$

3.6 Non-uniqueness

The isometric solution is optimum among cardinality D solutions.

Proposition 5 $\ell(X_S) = D, |S| = D \Leftrightarrow X_S X_S^T = I_D$

However, there exist Isometry Pursuit solutions of minimum norm that are not isometries. For example, *Mercedes-Benz frame* matrix X achieves minimium norm.

Example 1 *Let* $D = 2, P = 3$. *Let* $X_S = \begin{bmatrix} 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}, \beta = \begin{bmatrix} 0 & \frac{2}{3} \\ \frac{1}{\sqrt{3}} & -\frac{1}{3} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{3} \end{bmatrix}$ *Then* $X\beta = I_D$ *and* $\|\beta\|_{1,2} = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = 2 = D$. *Since the identity matrix is a solution regardless of* X_{-S}, *this is a non-isometric minimum loss solution.*

The upshot of this example is that IsometryPursuit can select solutions that are relatively non-isometric. The Mercedes-Benz frame is an example of a tight frame - a frame whose non-zero singular values are equal. While there are non-tight frames that have minimizing solutions, we do not here resolve whether there exist tight frames with unit norm columns that have minimum norm solutions.

3.7 Computation complexity

IsometryPursuit is more computationally efficient than BruteSearch search for the minimizer of ℓ. Interestingly, their asymptotic per-iteration complexities are the same.

Proposition 6 *The computational complexity* C_ℓ *of 5 is* $O(D^2 P)$.

The above proposition follows immediately from the computational complexity of singular value decomposition (**?**). The per-iteration complexity of IsometryPursuit is also limited by the computational complexity of singular value decomposition, albeit in the more intricate manner described in Section **??**.

Proposition 7 *The computational complexity of each step of Isometry Pursuit is* $O(D^2 P)$.

The difference therefore lies in the relation of the number of iterations required to achieve a given error tolerance with IsometryPursuit and BruteSearch. To optimize, we use the SCS interior point solver (**?**) from CVXPY (**??**), which is able to push sparse values arbitrarily close to 0 (**?**). SCS (aka ADMM) has asymptotic (in error) accuracy $\epsilon < O(\frac{1}{\mathcal{S}})$, where $\mathcal{S}$ is the number of iterations (**?**). This means that the number of iterations required to achieve a given error tolerance scales inversely with the desired level of accuracy. We examine the effect of iteration number on the convergence to a sparse solution in Section **??**.